

Expected Interview Questions & STAR Answers

Q1 (Technical): How do you address context window limits and retrieval latency in large-scale RAG systems?

STAR Answer:

- **Situation:** At Apex Intelligence Labs, enterprise search required querying 500k+ documents with latency under 50ms.
- **Task:** Architect a hybrid dense + sparse vector retrieval pipeline.
- **Action:** Implemented BM25 keyword filtering combined with ChromaDB dense embeddings, optimized with dynamic batching.
- **Result:** Reduced response latency by 68% while handling 5,000 req/sec.

Q2 (Behavioral): Describe a situation where you had to trade off model accuracy for inference speed.

STAR Answer:

- **Situation:** EdgeQuant library development.
- **Task:** Compress 13B parameter LLMs for edge device deployment.
- **Action:** Applied INT8/4-bit quantization via TensorRT and ONNX.
- **Result:** Retained 98.4% FP16 accuracy while reducing memory footprint by 62%.